DS ASSIGNMENT 1

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1.Sum of two matrices.

PROGRAM:

```
#include <stdio.h>
int main()
{
    int a[20][20], b[20][20];
    int d[20][20], i, j, c=0, r=0;
    printf("Enter no of rows and columns = ");
    scanf("%d %d", &r, &c);
    printf("Enter Matrix 1 = ");
    for(i=0; i<r; i++)
    {
     for(j=0; j<c; j++)
        {
         scanf("%d", &a[i][j]);
        }
    }

    printf("Enter Matrix 2 = ");
    for(i=0; i<r; i++)
    {
        for(j=0; j<c; j++)
        {
            scanf("%d", &b[i][j]);
        }
    }

    for(i=0; i<r; i++)
    {
        for(j=0; j<c; j++)
        {
            d[i][j] = a[i][j] + b[i][j];
        }
    }

    printf("In Sum = ");
    for(i=0; i<r; i++)
    {
        for(j=0; j<c; j++)
        {
            printf(" %d", d[i][j]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
Enter no of rows and columns = 2 2
Enter Matrix 1 = 1 2 3 4
Enter Matrix 2 = 1 2 3 4
In Sum =  2 4
 6 8
```

2. Write a C program to read n numbers into an array and display them in reverse order.

PROGRAM:

```
#include <stdio.h>
int main()
{
    int a[20], i;
    int n;

    printf("Enter n = ");
    scanf("%d", &n);

    printf("Enter elements = ");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    printf("\nReverse order = ");
    for(i = n - 1; i >= 0; i--)
    {
        printf("%d ", a[i]);
    }

    return 0;
}
```

OUTPUT:

```
Enter n = 4
Enter elements = 1 2 3 4

Reverse order = 4 3 2 1
```

3. Count the total number of duplicate elements in an array.

PROGRAM:

```
#include <stdio.h>
int main()
{
    int a[20], i, n, count = 0, j;
    printf("Enter n = ");
    scanf("%d", &n);
    printf("Enter elements = ");
    for(i = 0; i < n; i++)
```

```
    {
        scanf("%d", &a[i]);
    }

    for(i = 0; i < n; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(a[i] == a[j])
                count++;
        }
    }
    printf("No of duplicate elements = %d", count);
    return 0;
}
```

OUTPUT:

```
Enter n = 5
Enter elements = 1 2 3 2 1
No of duplicate elements = 2
```

4.Print unique elements in an array.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int a[50], i, j;
    int n, count;

    printf("Enter n= ");
    scanf("%d", &n);

    printf("Enter elements = ");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    for(i = 0; i < n; i++)
    {
        count = 0;

        for(j = 0; j < n; j++)
        {
            if(i != j && a[i] == a[j])
            {
                count++;
            }
        }

        if(count == 0)
        {
```

```
            printf("%d ", a[i]);
        }
    }

    return 0;
}
```

OUTPUT:

```
Enter n= 4
Enter elements = 1 1 2 3
2 3
```

5. Insert an element in a sorted array.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int a[20], i, n, item;
    printf("\nEnter n = ");
    scanf("%d", &n);
    printf("\nEnter %d elements: ", n);
    for(i = 0; i < n; i++)
        scanf("%d", &a[i]);
    printf("\nEnter number to be inserted = ");
    scanf("%d", &item);

    i = n - 1;

    while(i >= 0 && a[i] > item)
    {
        a[i + 1] = a[i];
        i--;
    }

    a[i + 1] = item;
    n++;

    printf("\nAfter insertion = ");
    for(i = 0; i < n; i++)
        printf("%d ", a[i]);

    return 0;
}
```

OUTPUT:

```
Enter n = 5

Enter 5 elements: 2 3 4 5 7

Enter number to be inserted = 6

After insertion = 2 3 4 5 6 7
```

6. Insert an element in an unsorted array.

PROGRAM:

```
#include <stdio.h>
int main()
{
    int a[100], i, n, pos = 0,item;

    printf("Enter n = ");
    scanf("%d", &n);

    printf("Enter array = ");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &a[i]);
    }

    printf("Enter position = ");
    scanf("%d", &pos);

    printf("\nEnter number to be inserted = ");
    scanf("%d", &item);

    printf("\nEnter position = ");
    scanf("%d", &pos);

    for(i = n; i > pos; i--)
    {
        a[i] = a[i - 1];
    }

    a[pos] = item;
    n++;

    printf("\nAfter insertion = ");
    for(i = 0; i < n; i++)
    {
        printf("%d ", a[i]);
    }

    return 0;
}
```

OUTPUT:

```
Enter n = 4
Enter array = 1 2 3 4

Enter number to be inserted = 0

Enter position = 0

After insertion = 0 1 2 3 4
```

7.Matrix multiplication.

PROGRAM:

```
#include <stdio.h>
int main()
{
    int a[10][10], b[10][10], d[10][10];
    int r1, c1, r2, c2;
    int i, j, k;

    printf("Enter rows & columns of matrix 1 = ");
    scanf("%d %d", &r1, &c1);

    printf("Enter rows & columns of matrix 2 = ");
    scanf("%d %d", &r2, &c2);

    if (c1 != r2)
    {
        printf("\nMultiplication Not possible");
    }
    else
    {
        printf("Enter Matrix 1 =\n");

        for (i = 0; i < r1; i++)
        {
            for (j = 0; j < c1; j++)
            {
                scanf("%d", &a[i][j]);
            }
        }

        printf("\nEnter Matrix 2 =\n");

        for (i = 0; i < r2; i++)
        {
            for (j = 0; j < c2; j++)
            {
                scanf("%d", &b[i][j]);
            }
        }

        printf("\nMatrix Multiplication =\n");

        for (i = 0; i < r1; i++)
        {
            for (j = 0; j < c2; j++)
            {
                d[i][j] = 0;

                for (k = 0; k < c1; k++)
                {
                    d[i][j] += a[i][k] * b[k][j];
                }
            }
        }

        for (i = 0; i < r1; i++)
```

```
        {
            for (j = 0; j < c2; j++)
            {
                printf("\t%d", d[i][j]);
            }
            printf("\n");
        }
    }

    return 0;
}
```

OUTPUT:

```
Enter rows & columns of matrix 1 = 2 3
Enter rows & columns of matrix 2 = 3 2
Enter Matrix 1 =
1 2 3 4 5 5 6

Enter Matrix 2 =
1 2 3 4 5 6

Matrix Multiplication =
        22      22
        54      44
```

8. Matrix transpose.

PROGRAM:

```
#include <stdio.h>
int main()
{
    int i, r, c, j;
    int a[20][20];

    printf("\nEnter rows & cols = ");
    scanf("%d %d", &r, &c);

    printf("\nEnter Matrix = ");
    for (i = 0; i < r; i++)
    {
        for (j = 0; j < c; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("\nTranspose Matrix = \n");
    for (i = 0; i < c; i++)
    {
        for (j = 0; j < r; j++)
        {
```

```
            printf("%d ", a[j][i]);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
Enter rows & cols = 2 3

Enter Matrix = 1 2 3 4 5 6

Transpose Matrix =
1 4
2 5
3 6
```

9.Right diagonal sum of a matrix.

PROGRAM:

```
#include <stdio.h>
int main()
{
    int i, n, j, sum = 0;
    int a[100][100];

    printf("Enter order = ");
    scanf("%d", &n);

    printf("\nEnter Matrix = ");

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    printf("\nRight Diagonal Sum = ");

    for(i = 0; i < n; i++)
    {
        sum += a[i][n - 1 - i];
    }

    printf("%d", sum);

    return 0;
}
```

OUTPUT:

```
Enter order = 3

Enter Matrix = 1 2 3 4 5 6 7 8 9

Right Diagonal Sum = 15
```

10. Check whether a matrix is an identity matrix.

PROGRAM:

```
#include <stdio.h>

int main()
{
    int i, j, n;
    int a[20][20];
    int identity = 1;

    printf("Enter order: ");
    scanf("%d", &n);

    printf("Enter matrix:\n");

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            scanf("%d", &a[i][j]);
        }
    }

    for(i = 0; i < n; i++)
    {
        for(j = 0; j < n; j++)
        {
            if(i == j)
            {
                if(a[i][j] != 1)
                    identity = 0;
            }
            else
            {
                if(a[i][j] != 0)
                    identity = 0;
            }
        }
    }

    if(identity == 1)
        printf("It is an Identity Matrix");
    else
        printf("It is not an Identity Matrix");

    return 0;
}
```

OUTPUT:

```
Enter order: 3
Enter matrix:
1 0 0 0 1 0 0 0 1
It is an Identity Matrix
```

11. Find the number of vowels, consonants, digits, and white spaces in a string.

PROGRAM:

```
#include <stdio.h>
#include <string.h>
int main()
{
    char a[100];
    int Vowel = 0, Conso = 0, Space = 0, Digit = 0;

    printf("Enter a string = ");
    fgets(a, sizeof(a), stdin);

    int len = strlen(a);

    for (int i = 0; i < len; i++)
    {
        if ((a[i] >= 'A' && a[i] <= 'Z') &&
            (a[i] >= 'a' && a[i] <= 'z'))
        {
            if (a[i] == 'a' || a[i] == 'e' || a[i] == 'i' ||
                a[i] == 'o' || a[i] == 'u' || a[i] == 'A' ||
                a[i] == 'E' || a[i] == 'I' || a[i] == 'O' ||
                a[i] == 'U')
            {
                Vowel++;
            }
            else
            {
                Conso++;
            }
        }
        else if (a[i] >= '0' && a[i] <= '9')
        {
            Digit++;
        }
        else if (a[i] == ' ')
        {
            Space++;
        }
    }

    printf("\nVowel = %d\nConsonant = %d\nDigits = %d\nWhite Space =
%d\n",
           Vowel, Conso, Digit, Space);

    return 0;
}
```

OUTPUT:

```
Enter a string: asmi 14 years 56 heigh
Vowels = 6
Consonants = 8
Digits = 4
White spaces = 4
```

12. Find the frequency of a character in a string.

PROGRAM:

```
#include <stdio.h>
int main()
{
    char a[50];
    int b[50], count = 0;
    int i, j;

    printf("Enter a string = ");
    scanf("%s", a);

    for(i = 0; a[i] != '\0'; i++)
    {
        count = 1;

        if(a[i] != ' ')
        {
            for(j = i + 1; a[j] != '\0'; j++)
            {
                if(a[i] == a[j])
                {
                    count++;
                    a[j] = ' ';
                }
            }

            b[i] = count;
        }
    }

    for(i = 0; a[i] != '\0'; i++)
    {
        if(a[i] != ' ')
        {
            printf("\n%c repeats %d times", a[i], b[i]);
        }
    }

    return 0;
}
```

OUTPUT:

```
Enter a string = array

a repeats 2 times
r repeats 2 times
y repeats 1 times
```

13. Count the number of words in a string.

PROGRAM:

```
#include <stdio.h>
#include <string.h>
int main()
{
    char a[100];
    int space = 0;

    printf("Enter a string = ");
    fgets(a, sizeof(a), stdin);

    for (int i = 0; a[i] != '\0'; i++)
    {
        if (a[i] == ' ' && a[i + 1] != '\0' && a[i + 1] != ' ')
        {
            space++;
        }
    }

    printf("No of words = %d", space + 1);

    return 0;
}
```

OUTPUT:

```
Enter a string = hello how are you
No of words = 4
```

14. Sort a string array.

PROGRAM:

```
#include <stdio.h>
#include <string.h>
int main()
{
    char a[5][20];
    char temp[20];
    int i, j;

    printf("Enter a string array:\n");

    for(i = 0; i < 5; i++)
    {
        scanf("%s", a[i]);
    }
```

```
    for(i = 0; i < 5; i++)
    {
        for(j = i + 1; j < 5; j++)
        {
            if(strcmp(a[i], a[j]) > 0)
            {
                strcpy(temp, a[i]);
                strcpy(a[i], a[j]);
                strcpy(a[j], temp);
            }
        }
    }

    printf("In Sorted array:\n");

    for(i = 0; i < 5; i++)
    {
        printf("%s\n", a[i]);
    }

    return 0;
}
```

OUTPUT:

```
Enter a string array:
asmi ram raju kalia moni
In Sorted array:
asmi
kalia
moni
raju
ram
```

15. Toggle the case of a sentence (lowercase to uppercase and vice versa.

PROGRAM:

```
#include <stdio.h>
#include <string.h>
void low_to_up(char a[])
{
    int i;
    for(i = 0; a[i] != '\0'; i++)
    {
        if(a[i] >= 'a' && a[i] <= 'z')
        {
            a[i] = a[i] - 32;
        }
    }
}

void up_to_low(char a[])
{
    int i;
    for(i = 0; a[i] != '\0'; i++)
    {
        if(a[i] >= 'A' && a[i] <= 'Z')
        {
```

```
            a[i] = a[i] + 32;
        }
    }
}

int main()
{
    char a[20];
    int n;

    printf("Enter a string = ");
    scanf("%s", a);

    printf("Enter 1 for uppercase to lower\n");
    printf("Enter 2 for lowercase to upper\n");
    scanf("%d", &n);

    switch(n)
    {
        case 1:
            up_to_low(a);
            break;

        case 2:
            low_to_up(a);
            break;
    }

    printf("\nAfter toggle\n");
    printf("%s", a);

    return 0;
}
```

OUTPUT:

```
Enter a string = kenni
Enter 1 for uppercase to lower
Enter 2 for lowercase to upper
2

After toggle
KENNI
```